

3.3 Badly Approximable Numbers

While Corollary 3.8 gives precise information about the behavior of typical real numbers, it does not say anything about the behavior of all real numbers. In this section we discuss a special class of real numbers that behave very differently to typical real numbers.

Definition 3.9. A real number $u = [a_1, a_2, \dots] \in (0, 1)$ is called *badly approximable* if there is some bound M with the property that $a_n \leq M$ for all $n \geq 1$.

Clearly a badly approximable number cannot satisfy (3.27). It follows that the set of all badly approximable numbers in $(0, 1)$ is a null set with respect to the Gauss measure, and hence is a null set with respect to Lebesgue measure⁽⁴³⁾. The next result explains the terminology: badly approximable numbers cannot be approximated very well by rationals.

Proposition 3.10. *A number $u \in (0, 1)$ is badly approximable if and only if there exists some $\varepsilon > 0$ with the property that*

$$\left| u - \frac{p}{q} \right| \geq \frac{\varepsilon}{q^2}$$

for all rational numbers $\frac{p}{q}$.

PROOF. If u is badly approximable, then (3.4) shows that

$$q_{n+1} \leq (M + 1)q_n$$

for all $n \geq 0$. For any q there is some n with $q \in (q_{n-1}, q_n]$, and by Proposition 3.3 and (3.15) we therefore have

$$\left| \frac{p}{q} - u \right| > \left| \frac{p_n}{q_n} - u \right| > \frac{1}{q_n q_{n+2}} > \frac{1}{(M + 1)^4 q^2}$$

as required.

Conversely, if

$$\left| u - \frac{p}{q} \right| \geq \frac{\varepsilon}{q^2}$$

for all rational numbers $\frac{p}{q}$ then, in particular,

$$\frac{\varepsilon}{q_n^2} \leq \left| u - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

by (3.13). This implies that

$$a_{n+1}q_n < a_{n+1}q_n + q_{n-1} = q_{n+1} < \frac{1}{\varepsilon}q_n,$$

so $a_{n+1} \leq \frac{1}{\varepsilon}$ for all $n \geq 1$. □

Example 3.11. Notice that $\frac{2}{\sqrt{5}-1} = \frac{\sqrt{5}+1}{2} \in (1, 2)$ and $\frac{\sqrt{5}+1}{2} - 1 = \frac{\sqrt{5}-1}{2}$. It follows that if

$$\frac{\sqrt{5}-1}{2} = [a_1, a_2, \dots]$$

then $a_1 + [a_2, a_3, \dots] \in (1, 2)$, so $a_1 = 1$, and hence

$$[a_2, a_3, \dots] = \frac{\sqrt{5}+1}{2} - 1 = \frac{\sqrt{5}-1}{2} = [a_1, a_2, \dots].$$

We deduce by the uniqueness of the continued fraction digits that

$$\frac{\sqrt{5}-1}{2} = [1, 1, 1, \dots],$$

so $\frac{\sqrt{5}-1}{2}$ is badly approximable.

Indeed, the specific number in Example 3.11 is, in a precise sense, the most badly approximable real number in $(0, 1)$. In the next section we generalize this example to show that all quadratic irrationals are badly approximable.

3.3.1 Lagrange's Theorem

The periodicity of the continued fraction expansion seen in Example 3.11 is a general property of quadratics. A real number u is called a *quadratic irrational* if $u \notin \mathbb{Q}$ and there are integers a, b, c with $au^2 + bu + c = 0$. Notice that u is a quadratic irrational if and only if $\mathbb{Q}(u)$ is a subfield of $\mathbb{R}$ of degree 2 over $\mathbb{Q}$.

Definition 3.12. A continued fraction $[a_0; a_1, \dots]$ is *eventually periodic* if there are numbers $N \geq 0$ and $k \geq 1$ with $a_{n+k} = a_n$ for all $n \geq N$. Such a continued fraction will be written

$$[a_0; a_1, \dots, a_{N-1}, \overline{a_N, \dots, a_{N+k}}].$$

The main result describing the special properties of quadratic irrationals is Lagrange's Theorem [218, Sect. 34].

Theorem 3.13 (Lagrange). *Let u be an irrational positive real number. Then the continued fraction expansion of u is eventually periodic if and only if u is a quadratic irrational.*

PROOF. Assume first that $u = [\overline{a_0; a_1, \dots, a_k}]$ has a strictly periodic continued fraction expansion, so that $u_{k+1} = u_0 = u$. Thus

$$u = \frac{up_k + p_{k-1}}{uq_k + q_{k-1}}$$

by (3.20), so

$$u^2 q_k + u(q_{k-1} - p_k) - p_{k-1} = 0$$

and u is a quadratic irrational (u cannot be rational, since it has an infinite continued fraction; alternatively notice that the quadratic equation satisfied by u has discriminant $(q_{k-1} - p_k)^2 + 4q_k p_{k-1} = (q_{k-1} + p_k)^2 - 4(-1)^k$ by (3.8), so cannot be a square).

Now assume that

$$u = [a_0; \dots, a_{N-1}, \overline{a_N, \dots, a_{N+k}}].$$

Then, by (3.20),

$$u = \frac{[\overline{a_N; a_{N+1}, \dots, a_{N+k}}]p_{N-1} + p_{N-2}}{[\overline{a_N; a_{N+1}, \dots, a_{N+k}}]q_{N-1} + q_{N-2}},$$

so $\mathbb{Q}(u) = \mathbb{Q}([\overline{a_N; a_{N+1}, \dots, a_{N+k}}])$, and therefore u is a quadratic irrational.

The converse is more involved⁽⁴⁴⁾. Assume now that u is a quadratic irrational, with

$$f_0(u) = \alpha_0 u^2 + \beta_0 u + \gamma_0 = 0$$

for some $\alpha_0, \beta_0, \gamma_0 \in \mathbb{Z}$ and $\delta = \beta_0^2 - 4\alpha_0\gamma_0$ not a square. We claim that for each $n \geq 0$ there is a polynomial

$$f_n(x) = \alpha_n x^2 + \beta_n x + \gamma_n$$

with

$$\beta_n^2 - 4\alpha_n\gamma_n = \delta$$

and with the property that $f_n(u_n) = 0$. This claim again follows from the fact that $\mathbb{Q}(u) = \mathbb{Q}(u_n)$, but we will need specific properties of the numbers $\alpha_n, \beta_n, \gamma_n$, so we proceed by induction.

Assume such a polynomial exists for some $n \geq 0$. Since $u_n = a_n + \frac{1}{u_{n+1}}$, we therefore have

$$u_{n+1}^2 f_n\left(a_n + \frac{1}{u_{n+1}}\right) = 0.$$

The resulting relation for u_{n+1} may be written in the form

$$f_{n+1}(x) = \alpha_{n+1} x^2 + \beta_{n+1} x + \gamma_{n+1}$$

where

$$\begin{aligned}\alpha_{n+1} &= a_n^2 \alpha_n + a_n \beta_n + \gamma_n, \\ \beta_{n+1} &= 2a_n \alpha_n + \beta_n,\end{aligned}\tag{3.30}$$

$$\gamma_{n+1} = \alpha_n.\tag{3.31}$$

It is clear that $\alpha_{n+1}, \beta_{n+1}, \gamma_{n+1} \in \mathbb{Z}$, and a simple calculation shows that

$$\beta_{n+1}^2 - 4\alpha_{n+1}\gamma_{n+1} = \beta_n^2 - 4\alpha_n\gamma_n,$$

proving the claim.

Notice that all the polynomials f_n have the same discriminant δ , which is not a square, so $\alpha_n \neq 0$ for $n \geq 0$. If there is some N with $\alpha_n > 0$ for all $n \geq N$, then (3.30) shows that the sequence $\beta_N, \beta_{N+1}, \dots$ is increasing since $a_n > 0$ for $n \geq 1$. Thus for large enough n , by (3.31), all three of α_n, β_n and γ_n are positive. This is impossible, since $f_n(u_n) = 0$ and $u_n > 0$. A similar argument shows that there is no N with $\alpha_n < 0$ for all $n \geq N$. We deduce that α_n must change in sign infinitely often, so in particular there is an infinite set $A \subseteq \mathbb{N}$ with the property that $\alpha_n \alpha_{n-1} < 0$ for all $n \in A$. By (3.31), it follows that $\alpha_n \gamma_n < 0$ for all $n \in A$. Now $\beta_n^2 - 4\alpha_n \gamma_n = \delta$, so for $n \in A$ we must have

$$\begin{aligned}|\alpha_n| &\leq \frac{1}{4}\delta, \\ |\beta_n| &< \sqrt{\delta},\end{aligned}$$

and

$$|\gamma_n| \leq \frac{1}{4}\delta.$$

It follows that as n runs through the infinite set A there are only finitely many possibilities for the polynomials f_n , so there must be some $n_0 < n_1 < n_2$ with $f_{n_0} = f_{n_1} = f_{n_2}$. Since a quadratic polynomial has only two zeros, and $u_{n_0}, u_{n_1}, u_{n_2}$ are all zeros of the same polynomial, we see that two of them coincide so the continued fraction expansion of u is eventually periodic. $\square$

Corollary 3.14. *Any quadratic irrational is badly approximable.*

PROOF. This is an immediate consequence of Theorem 3.13 and Definition 3.9. $\square$

It is not known if any other algebraic numbers are badly approximable.

Exercises for Sect. 3.3

Exercise 3.3.1. ⁽⁴⁵⁾ Show that $\mathbb{Q}(\sqrt{5})$ contains infinitely many elements with a uniform bound on their partial quotients, by checking that the numbers $[1^{k+1}, 4, 2, 1^k, 3]$ for $k \geq 0$ all lie in $\mathbb{Q}(\sqrt{5})$ (here 1^k denotes the string